

Lab 7

Incarceration, Arrests, & Poverty

with data from the ACS, Vera Institute, San Francisco, & Chicago

DATA 20195 Spring 2026

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History of Poverty Thresholds

- President Lyndon B. Johnson announced the **‘War on Poverty’** during his State of the Union Address in January 1964
- Mollie Orshansky – **economist** and **statistician** who developed the **Orshansky Poverty Thresholds**
 - worked with **USDA** from 1953 to 1958, directing the collection and analysis of **data on food consumption** and **expenditures of American households**



Image Credits: Wikipedia

TABLE 13.—Quantities of food groups in food plans per capita per week

Food budget by USDA, 1963	Food group	Food plans				
		Low-cost	Another low-cost ¹	Economy (cheapest)	Moderate-cost	Liberal
	Milk, cheese, ice cream ²quarts..	4.6	4.4	4.1	4.7	5.0
	Meat, poultry, fish.....pounds..	2.6	2.2	1.6	4.1	4.7
	Eggs.....dozen..	.5	.5	.4	.5	.6
	Dry beans, peas, nuts.....pounds..	.3	.4	.4	.2	.1
	Flour, cereals, baked goods ³do..	2.9	3.3	3.2	2.6	2.4
	Citrus fruit, tomatoes.....do..	2.1	1.9	1.7	2.5	2.9
	Dark-green and deep-yellow vegetables.....do..	.7	.7	.7	.7	.7
	Potatoes.....do..	2.1	1.6	2.8	1.9	1.7
	Other vegetables and fruits.....do..	4.4	4.1	2.8	5.1	5.6
	Fats, oils.....do..	.5	.6	.6	.6	.6
	Sugars, sweets.....do..	.6	.6	.7	.8	1.0

¹ Based on population weights of 1960 census.

² Particularly suitable for food habits of families in the Southeastern States.

³ Fluid whole milk or its equivalent in cheese, evaporated milk, dry milk, or ice cream.

⁴ Weight in terms of flour and cereal.

Found 1/3 after tax income spent on food

$$\text{Poverty threshold} = 3 * \$$$

Mollie wanted the ‘low-cost’ budget to be used for the threshold!

Poverty Thresholds today

- Census Bureau uses a set of money income thresholds **that vary by family size & composition** to determine poverty, **evaluated on the total pre-tax income of all working family members**
 - do not vary geographically
 - does not include account for non-cash benefits (public housing, Medicaid, and food stamps)

Poverty Thresholds for 2025 by Size of Family and Number of Related Children Under 18 Years (in \$)

Source: U.S. Census Bureau, 2026.

Size of family unit	Related children under 18 years								
	None	One	Two	Three	Four	Five	Six	Seven	Eight or more
One person (unrelated individual):									
Under 65 years.....	16,749								
65 years and over.....	15,440								
Two people:									
Householder under 65 years.....	21,558	22,190							
Householder 65 years and over.....	19,460	22,106							
Three people.....	25,183	25,913	25,938						
Four people.....	33,207	33,750	32,649	32,762					
Five people.....	40,045	40,628	39,384	38,421	37,833				
Six people.....	46,060	46,242	45,289	44,376	43,018	42,213			
Seven people.....	52,997	53,328	52,187	51,392	49,911	48,183	46,287		
Eight people.....	59,273	59,796	58,720	57,777	56,439	54,740	52,972	52,523	
Nine people or more.....	71,301	71,647	70,694	69,894	68,581	66,774	65,139	64,734	62,241

What are some limitations of these measures?

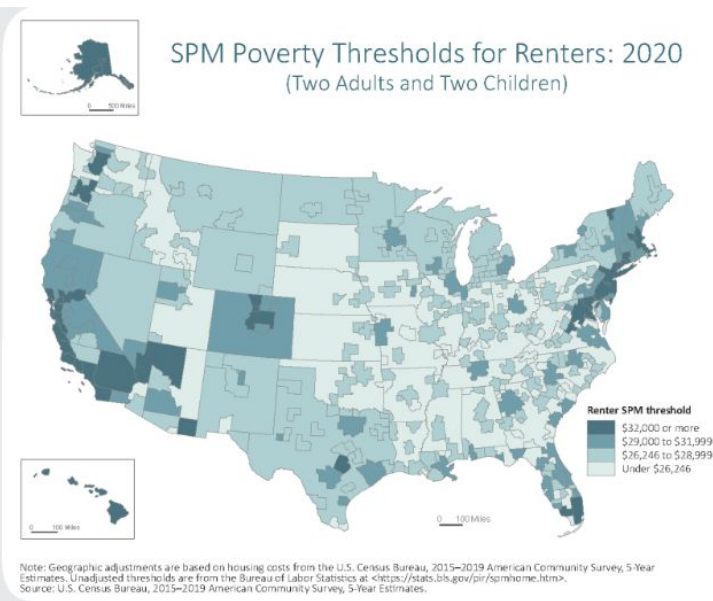
Official Poverty Measures (OPM) vs. Supplemental Poverty Measures (SPM)

- Census **started calculating the SPM in 2011**, but due to **sensitivity and granularity of data about a family required** to calculate the SPM, they only **report it at the state and national level**.
- Updates
 - **Expands definition of 'family'** to include co-resident unrelated children, foster children, and unmarried partners
 - Basic needs **not just a food budget**, expanded to include clothes, shelter, utilities, telecommunications
 - Varies geographically

Find out more!

https://www.census.gov/library/visualizations/2021/demo/poverty_measure-how.html

<https://www.irp.wisc.edu/resources/how-is-poverty-measured/>



Vera Institute Data

Organization

- Started with the **Manhattan Bail Project in 1961**
 - help defendants who were unable to pay the bonds in NYC
 - interviewed defendants to gather information on community ties
 - low risk individuals were recommended for release
- Now a national nonprofit research and policy organization focused on criminal justice reform

Data

- Geographic Identifiers (e.g. `fips`, `metro_area`)
- County Population statistics (e.g. `total_pop`, `male_pop_15to64`)
- Jail/Prison Population (e.g. `total_jail_pop`, `total_prison_pop`)
- Jail/Prison Population Rates (e.g. `total_jail_pop_rate`, `total_prison_pop_rate`)

Discussion: why use the age range 15–64?

The **Vera Institute** has a policy of only publishing incarceration data for persons aged 15–64 since it gives a “**more accurate picture of incarceration rates**” and allows for better comparison across counties.

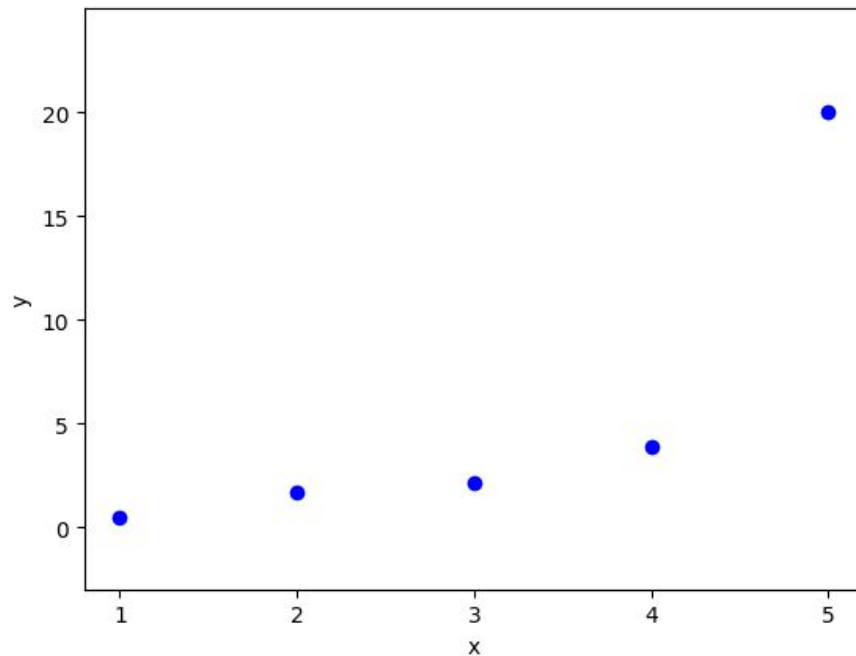
How could widening the age range reduce the accuracy of the collected data?

Would this impact how trustworthy a county-to-county comparison is?

To get a more accurate picture of incarceration rates, Vera removed people under the age of 15 and over 64 from the general population for the purposes of calculating incarceration and admission rates, since these groups are at very low risk of incarceration. Because the proportion of these groups varies greatly by county, keeping them in would skew rates and make comparisons between counties difficult. Note that this method differs from most other calculations of statewide and national incarceration rates, which use either the total resident population or the population aged 18 and older.¹⁰

Huber Regression

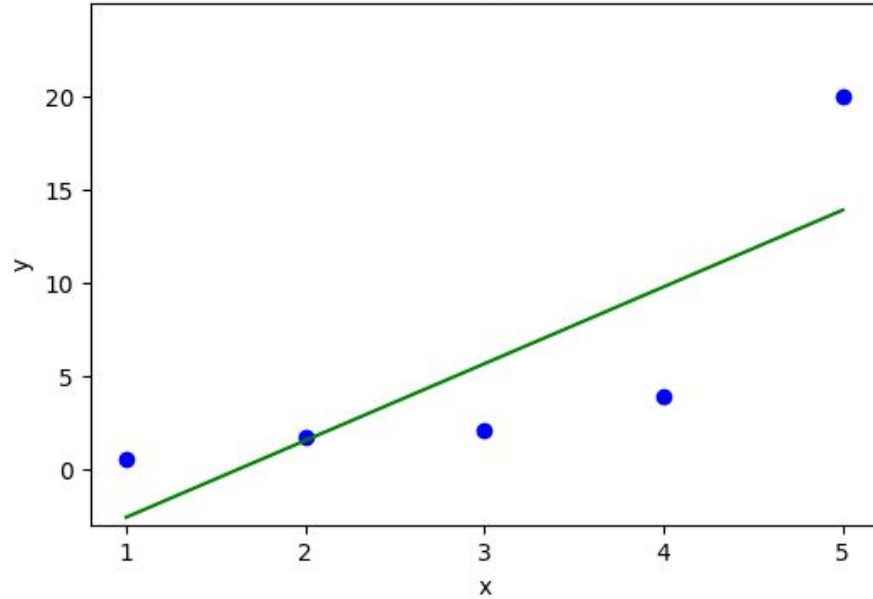
Suppose we have a dataset with an outlier.



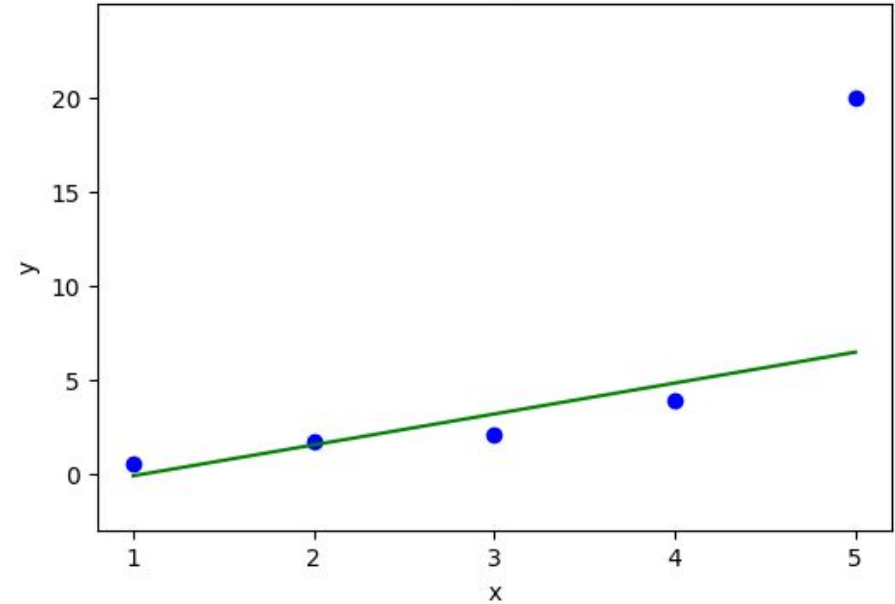
What would the regression line look like here?

Huber Regression

Linear Regression



Huber Regression



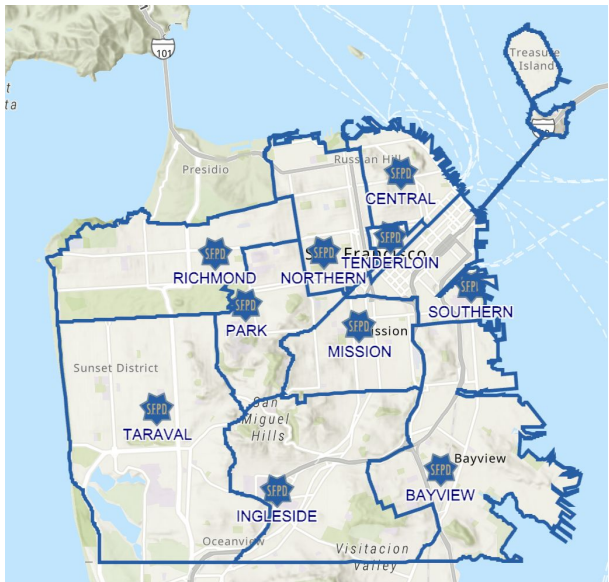
How would you draw the line differently?

San Francisco Incidents Dataset

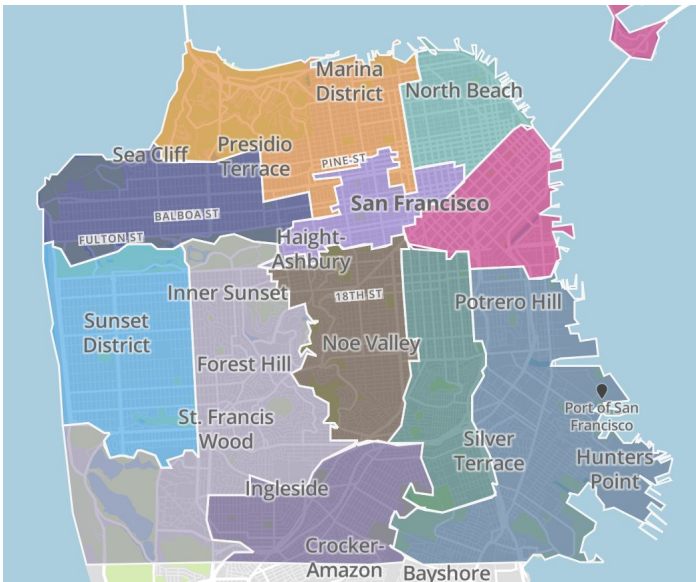
	incident_datetime	report_datetime	incident_category	incident_subcategory	incident_description	intersection	police_district	neighborhood	supervisor_district	lat	lon
0	2021-07-07 08:18:00	2021-07-07 08:28:00	Assault	Simple Assault	Battery	23RD ST \ MISSION ST	Mission	Mission	9.0	37.753837	-122.418594
1	2021-06-04 09:40:00	2021-06-04 09:42:00	Assault	Simple Assault	Battery	GEARY ST \ POLK ST	Northern	Tenderloin	5.0	37.785893	-122.419739
2	2022-05-07 21:02:00	2022-05-07 21:25:00	Assault	Simple Assault	Battery	MASON ST \ DERBY ST	Central	Tenderloin	3.0	37.787429	-122.409931
3	2021-09-30 09:00:00	2021-09-30 09:22:00	Assault	Simple Assault	Battery	16TH ST \ MISSION ST	Mission	Mission	9.0	37.765051	-122.419669
4	2021-12-18 00:45:00	2021-12-18 00:45:00	Other Miscellaneous	Other	Investigative Detention	POLK ST \ FERN ST	Northern	Nob Hill	3.0	37.788223	-122.420210

All report locations are mapped to the nearest intersection for anonymity.

SF Police Districts



SF Board of Supervisors Districts



★The Board of Supervisors is the legislative body for SF.

- Current Supervisor Districts
- SUPERVISORIAL DISTRICT 1
 - SUPERVISORIAL DISTRICT 2
 - SUPERVISORIAL DISTRICT 3
 - SUPERVISORIAL DISTRICT 4
 - SUPERVISORIAL DISTRICT 5
 - SUPERVISORIAL DISTRICT 6
 - SUPERVISORIAL DISTRICT 7
 - SUPERVISORIAL DISTRICT 8
 - SUPERVISORIAL DISTRICT 9
 - SUPERVISORIAL DISTRICT 10
 - SUPERVISORIAL DISTRICT 11

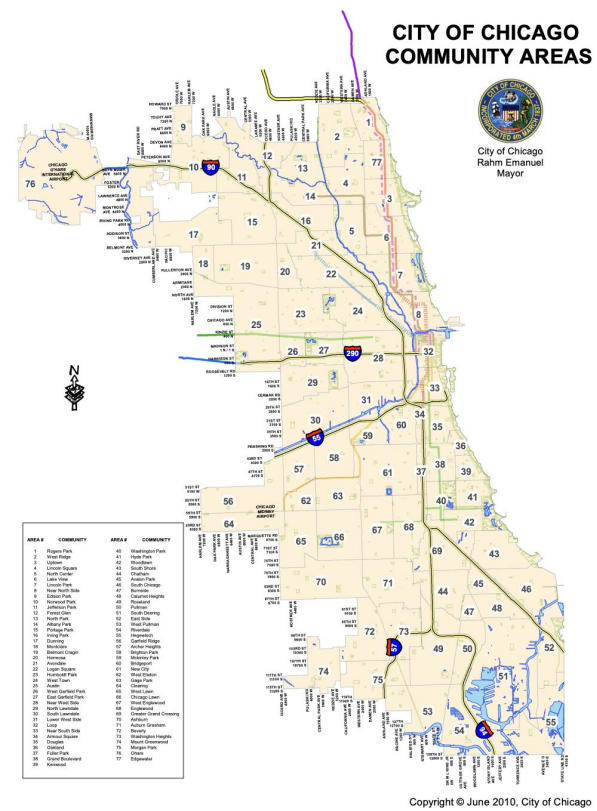
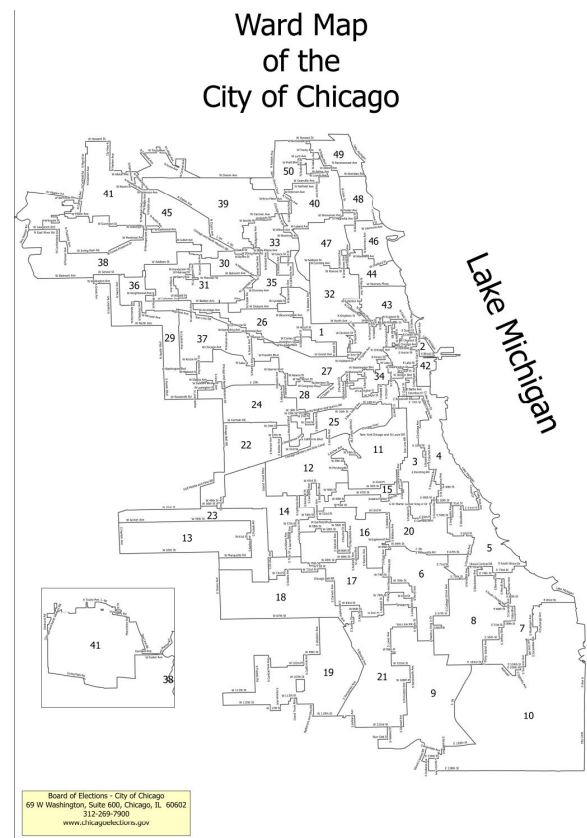
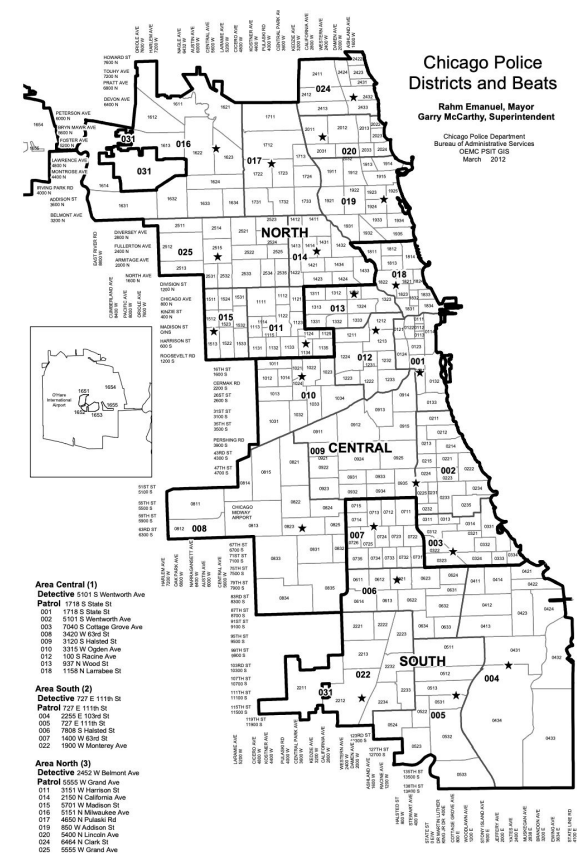
Chicago Arrest Dataset

	case_number	date	block	iucr	type	Description	location_type	is_domestic	beat	district	ward	community_area	fbi_code	updated_on	lat	lon
0	JC100006	2018-12-31 23:56:00	018XX S ALLPORT ST	0440	BATTERY	AGG: HANDS/FIST/FEET NO/MINOR INJURY	OTHER	0	1233	12	25	31	08B	2019-01-10 15:16:50	41.857068	-87.657625
1	JB574407	2018-12-31 23:44:00	047XX N RACINE AVE	1330	CRIMINAL TRESPASS	TO LAND	MOVIE HOUSE/THEATER	0	1913	19	46	3	26	2019-01-10 15:16:50	41.968463	-87.659670
2	JC100036	2018-12-31 23:43:00	071XX S VINCENNES AVE	143A	WEAPONS VIOLATION	UNLAWFUL POSS OF HANDGUN	GAS STATION	0	731	7	6	69	15	2019-01-10 15:16:50	41.764632	-87.628312
3	JC100004	2018-12-31 23:42:00	030XX N MARMORA AVE	2022	NARCOTICS	POSS: COCAINE	ALLEY	0	2514	25	30	19	18	2019-01-10 15:16:50	41.935640	-87.773689
4	JC100117	2018-12-31 23:40:00	088XX S COTTAGE GROVE AVE	1477	WEAPONS VIOLATION	RECKLESS FIREARM DISCHARGE	STREET	0	632	6	8	44	15	2019-01-10 15:16:50	41.733551	-87.604749

iucr, fbi_code: Crime type code (read more [here](#))

is_domestic: (Yes/no) the crime is domestic violence rated

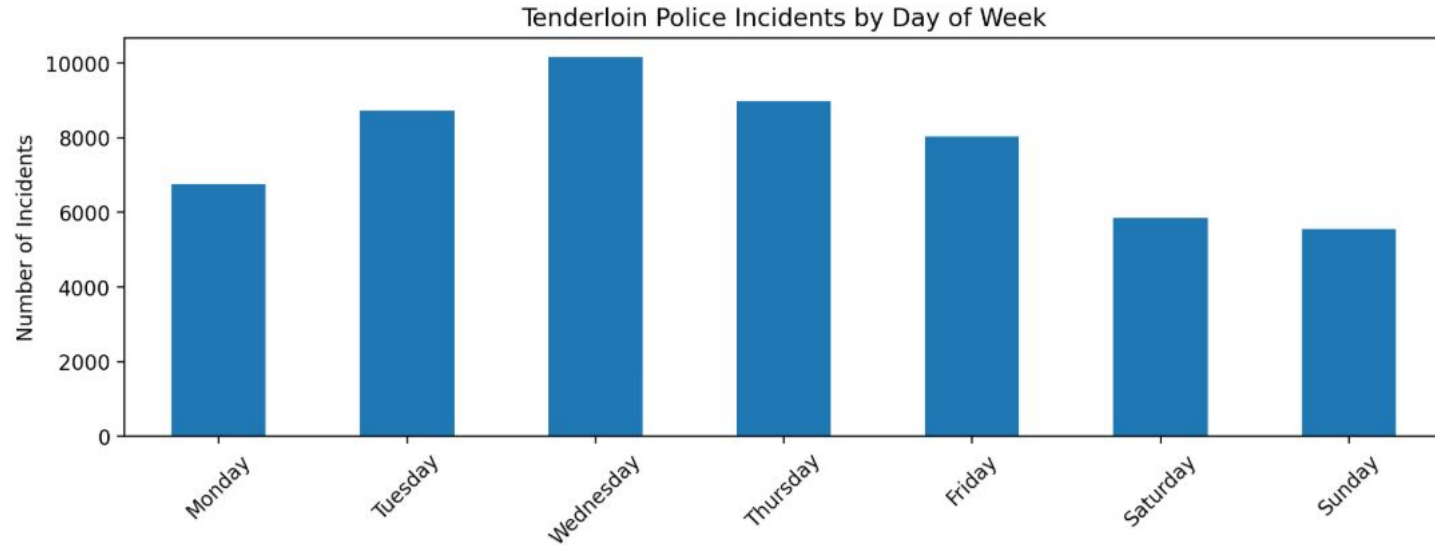
Chicago Arrest Dataset



Discussion: Wednesdays in the Tenderloin

There is a notable increase in the number of police incidents in the Tenderloin on Wednesdays

What could explain this?



Discussion: Comparing the Cities

We've set up dataframes to look at San Francisco and Chicago, but haven't done too much that is comparative.

Thinking back to how we handled the data in both cases...

What would be some steps towards a comparative analysis?

What datasets that we've previously seen in lab would you consider connecting to the crime information?

Which granularities of the crime data would help your analysis?

Which spatial operations (spatial joins, dissolve, buffers, network distances) would come in handy in your analysis?